

Business Operations and Sales Performance Analysis



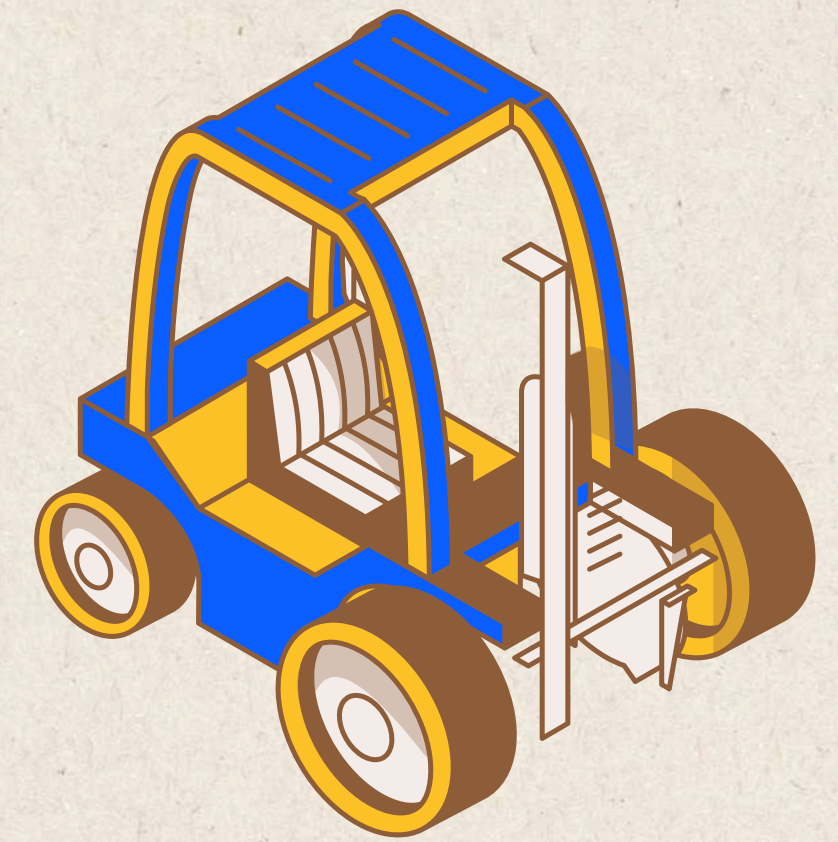
Presentation by
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Problem Statement

A manufacturing and retail company wants to analyze its overall business operations including production efficiency, supplier performance, logistics costs, and product sales. The data is stored across multiple tables (sales_data, supplier, manufacturing, logistics, and product_table).

The company seeks insights to:

1. Identify high-performing product types and top-selling products.
2. Evaluate supplier efficiency in terms of production volume and lead time.
3. Understand logistics costs across transportation modes.
4. Detect areas of improvement in manufacturing (defect rates, production costs).
5. Measure overall business revenue and profitability drivers.



Solution Approach

The project uses SQL to query and analyze key business metrics:

1. Sales Insights

- Display skincare product sales, including quantity and revenue.
- Identify products generating above-average revenue.
- Calculate total revenue across all products.
- Find total products sold by product type.

2. Supplier Analysis

- List all suppliers with locations.
- Identify top 2 suppliers by production volume.
- Find supplier(s) with the shortest lead time.

3. Manufacturing Analysis

- Find highest and lowest defect rates.
- Calculate total manufacturing cost per product type.

4. Logistics and Transportation

- Compute average and total shipping costs per transportation mode.
- Identify transportation modes with the highest total costs.
- Show products associated with the highest-cost transport modes.

5. Comprehensive Join Queries

- Combine supplier, logistics, and product details to understand supply chain links.
- Analyze product performance by joining sales_data and product_table.



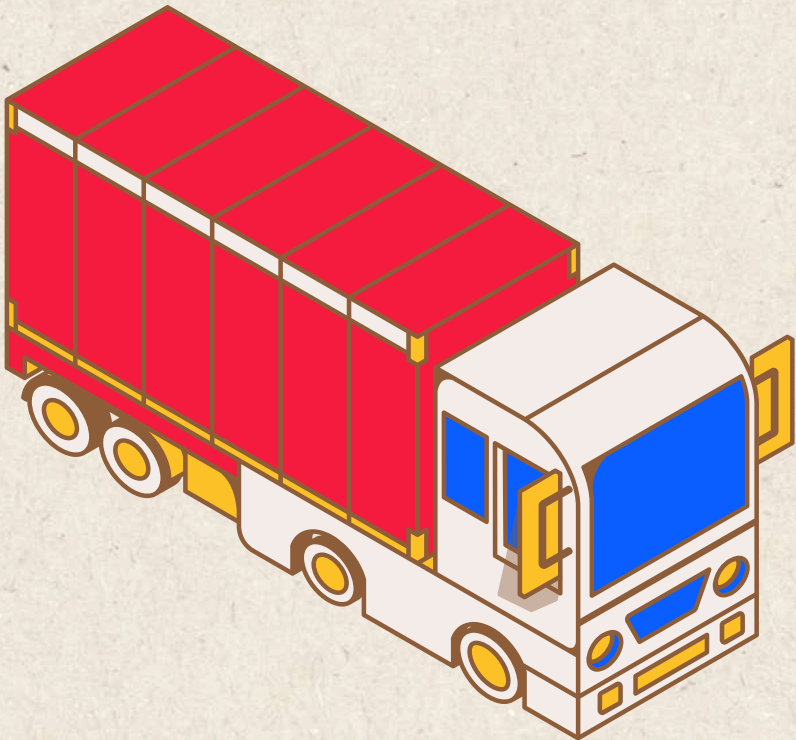
- Show sales of skincare products (SKU + number sold + revenue)

Code

```
-- Show sales of skincare products (SKU + number sold + revenue)
SELECT p.Product_type, s.SKU, s.Number_of_products_sold, s.Revenue_generated
FROM a1.sales_data s
JOIN a1.product_table p ON s.SKU = p.SKU
WHERE p.Product_type = 'skincare';
```

Output

	Product_type	SKU	Number_of_products_sold	Revenue_generated
▶	skincare	SKU1	736	7460.900065
	skincare	SKU3	83	7766.836426
	skincare	SKU4	871	2686.505152
	skincare	SKU6	65	7823.47656
	skincare	SKU9	980	4971.145988
	skincare	SKU10	996	2330.965802
	skincare	SKU11	960	6099.944116
	skincare	SKU12	840	4050.700410



- List all suppliers with their locations

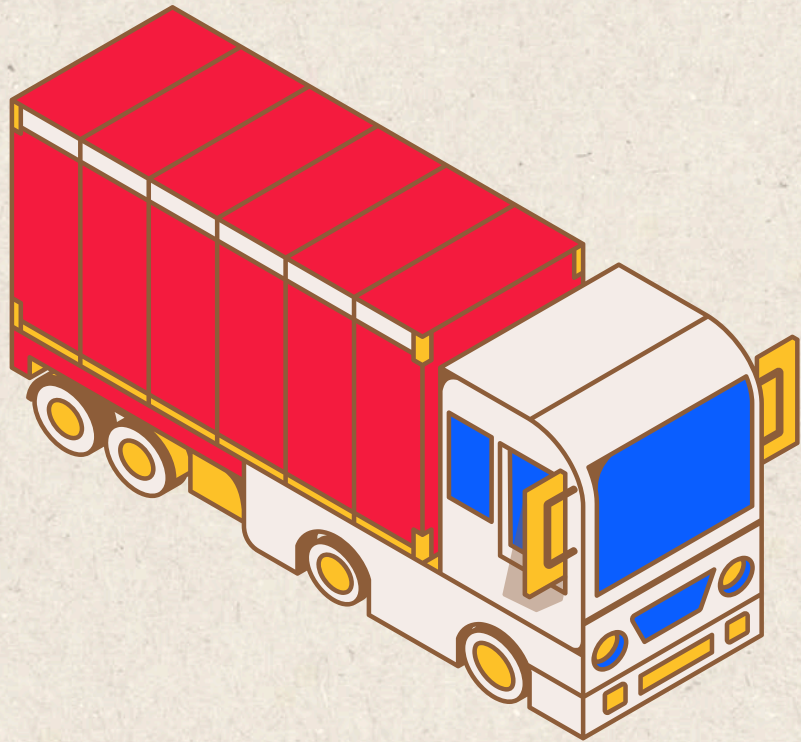
Code

```
-- List all suppliers with their locations

SELECT DISTINCT Supplier_name, Location
FROM a1.supplier;
```

Output

Result Grid			Filter Rows:
	Supplier_name	Location	
▶	Supplier 3	Mumbai	
	Supplier 1	Mumbai	
	Supplier 5	Kolkata	
	Supplier 1	Delhi	
	Supplier 4	Bangalore	
	Supplier 3	Kolkata	
	Supplier 4	Mumbai	
	Supplier 2	Chennai	
	Supplier 2	Kolkata	



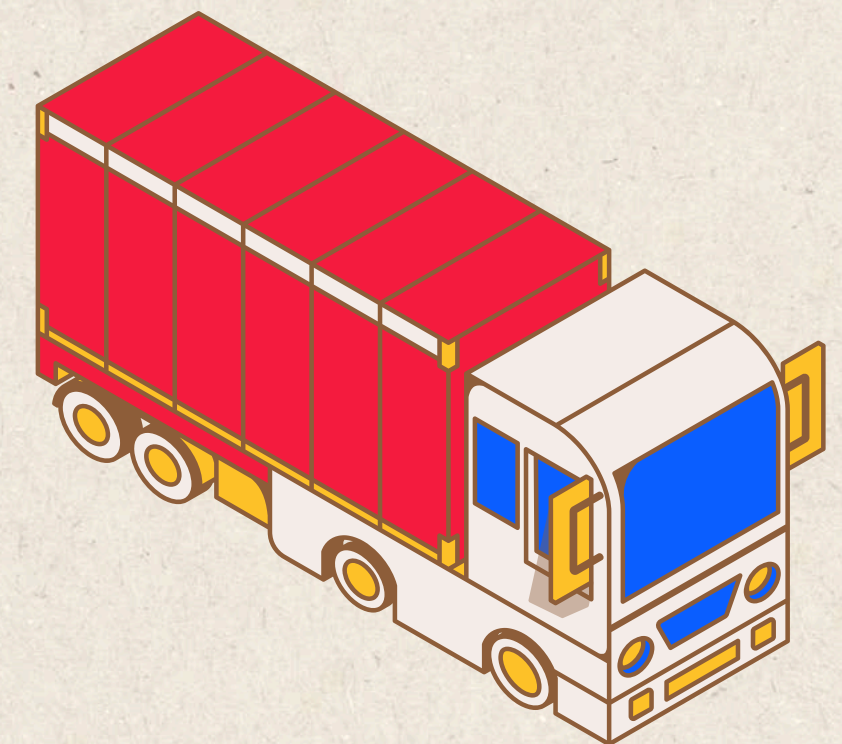
- Show total revenue generated from all products

Code

```
SELECT SUM(Revenue_generated) AS TotalRevenue  
FROM a1.sales_data;
```

Output

	TotalRevenue
▶	577604.81874



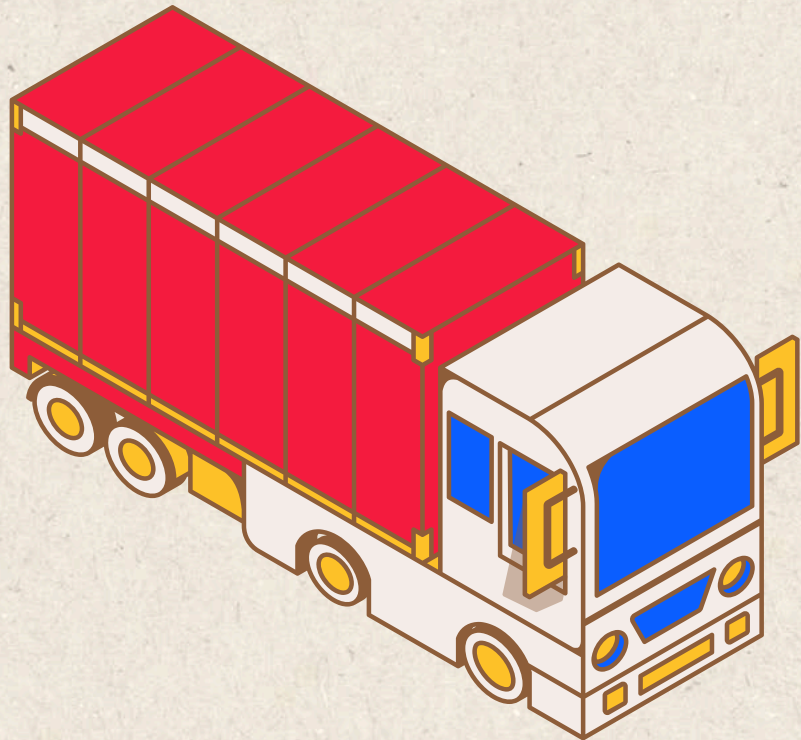
- Show sales of each product type

Code

```
SELECT p.Product_type, SUM(s.Number_of_products_sold) AS TotalSold
FROM a1.sales_data s
JOIN a1.product_table p ON s.SKU = p.SKU
GROUP BY p.Product_type;
```

Output

Result Grid		Filter Rows:
	Product_type	TotalSold
▶	hairecare	13611
	skincare	20731
	cosmetics	11757



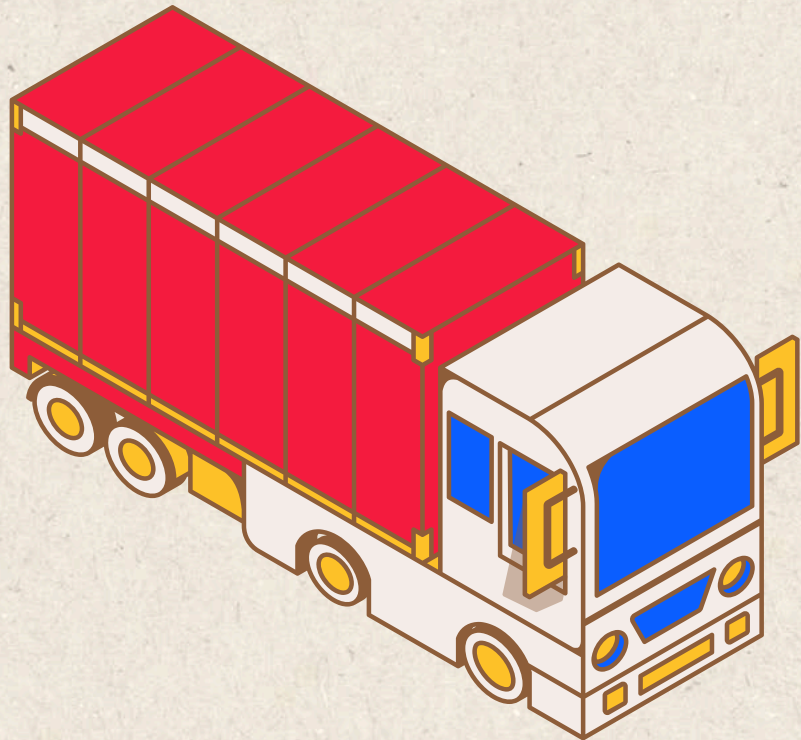
- Find the highest and lowest defect rate in manufacturing

Code

```
SELECT MAX(Defect_rates) AS MaxDefect, MIN(Defect_rates) AS MinDefect
FROM a1.manufacturing;
```

Output

Result Grid			Filter Rows:	
	MaxDefect	MinDefect		
▶	4.939255289	0.018607568		



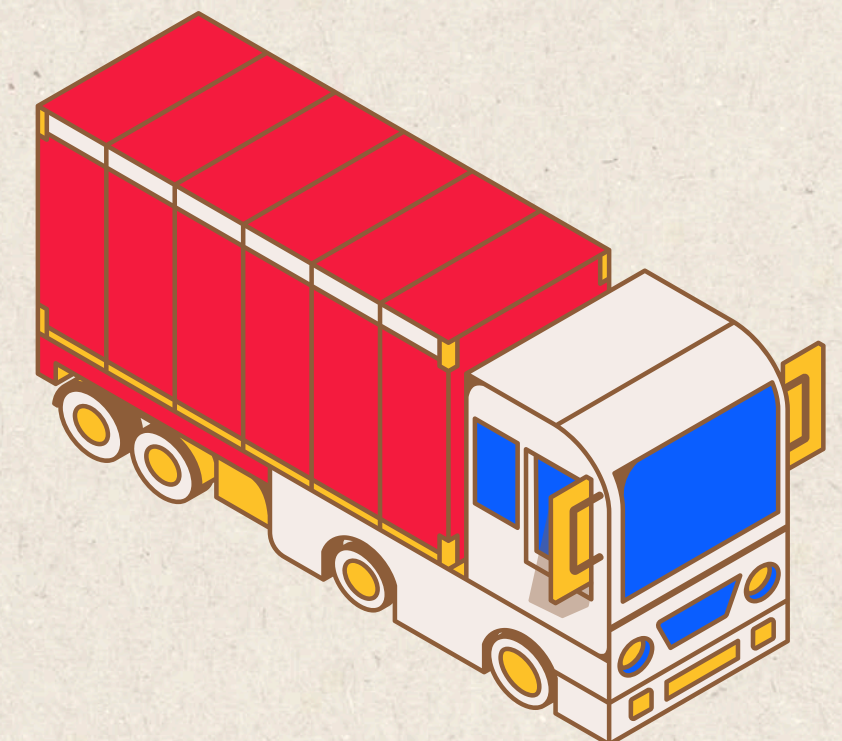
- Find top 2 suppliers by production volumes

Code

```
SELECT Supplier_name, SUM(Production_volumes) AS TotalVolume
FROM a1.supplier
GROUP BY Supplier_name
ORDER BY TotalVolume DESC
LIMIT 2;
```

Output

	Supplier_name	TotalVolume
▶	Supplier 2	14105
	Supplier 1	13545



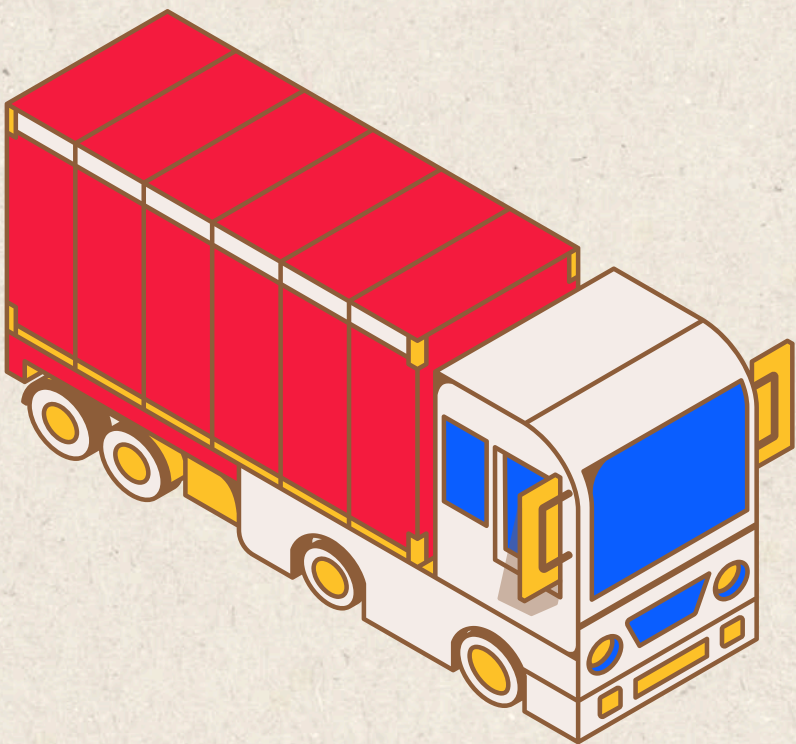
- Show average shipping cost by transportation mode

Code

```
SELECT Transportation_modes, AVG(Costs) AS AvgCost
FROM a1.logistics
GROUP BY Transportation_modes;
```

Output

	Transportation_modes	AvgCost
▶	Road	553.3859875482757
	Air	561.7125960615384
	Rail	541.7475556571429
	Sea	417.8191482647058



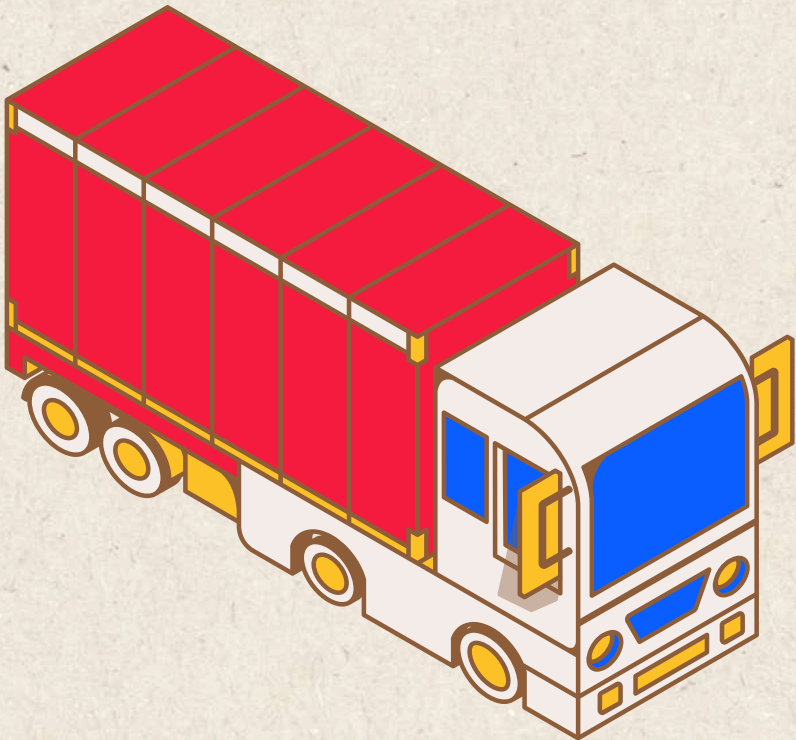
- Find products with revenue above average

Code

```
SELECT SKU, Revenue_generated
FROM a1.sales_data
WHERE Revenue_generated > (SELECT AVG(Revenue_generated) FROM a1.sales_data) limit 5;
```

Output

	SKU	Revenue_generated
▶	SKU0	8661.996792
	SKU1	7460.900065
	SKU2	9577.749626
	SKU3	7766.836426
	SKU6	7823.47656



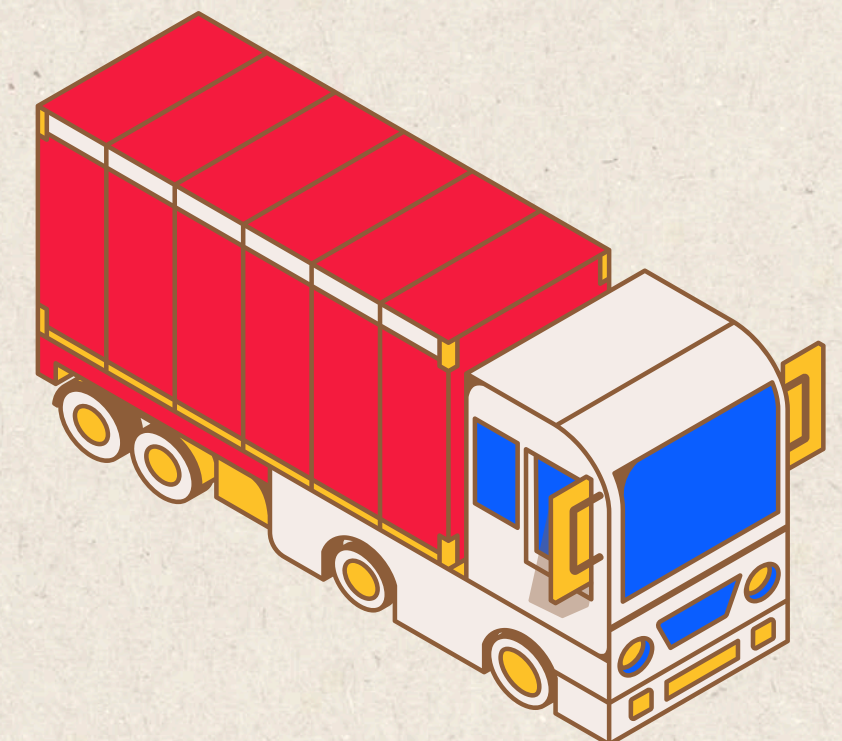
- Show supplier(s) with the shortest lead time

Code

```
SELECT Supplier_name, Lead_time  
FROM a1.supplier  
WHERE Lead_time = (SELECT MIN(Lead_time) FROM a1.supplier);
```

Output

	Supplier_name	Lead_time
▶	Supplier 5	1
	Supplier 5	1
	Supplier 1	1
	Supplier 4	1



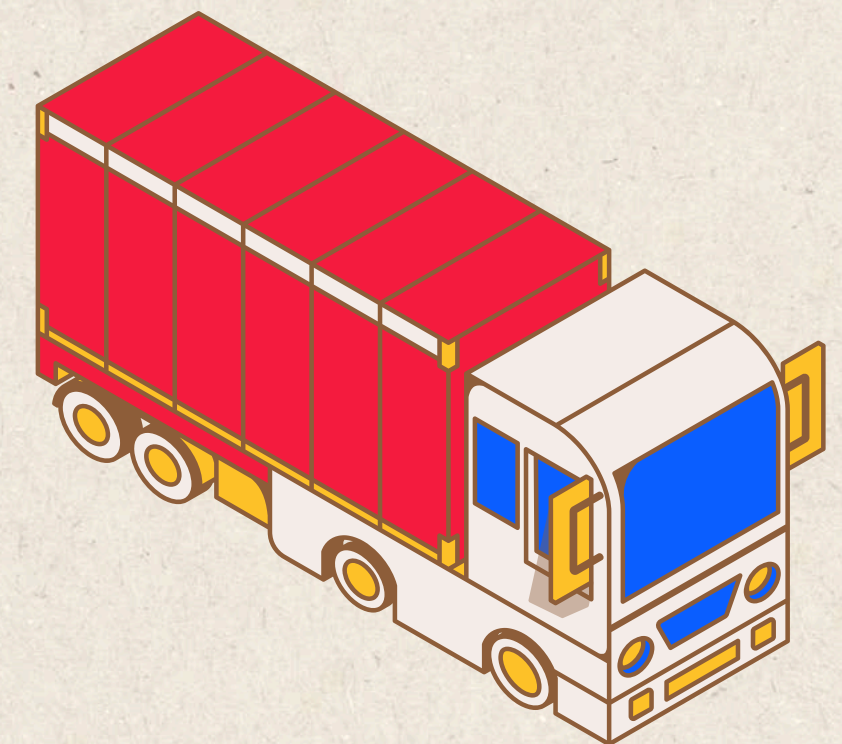
- Show total manufacturing cost per product type

Code

```
SELECT p.Product_type, SUM(m.Manufacturing_costs) AS TotalCost
FROM a1.product_table p
JOIN a1.manufacturing m ON p.SKU = m.SKU
GROUP BY p.Product_type;
```

Output

	Product_type	TotalCost
▶	haircare	1647.5717762899997
	skincare	1959.7262949529995
	cosmetics	1119.3712529



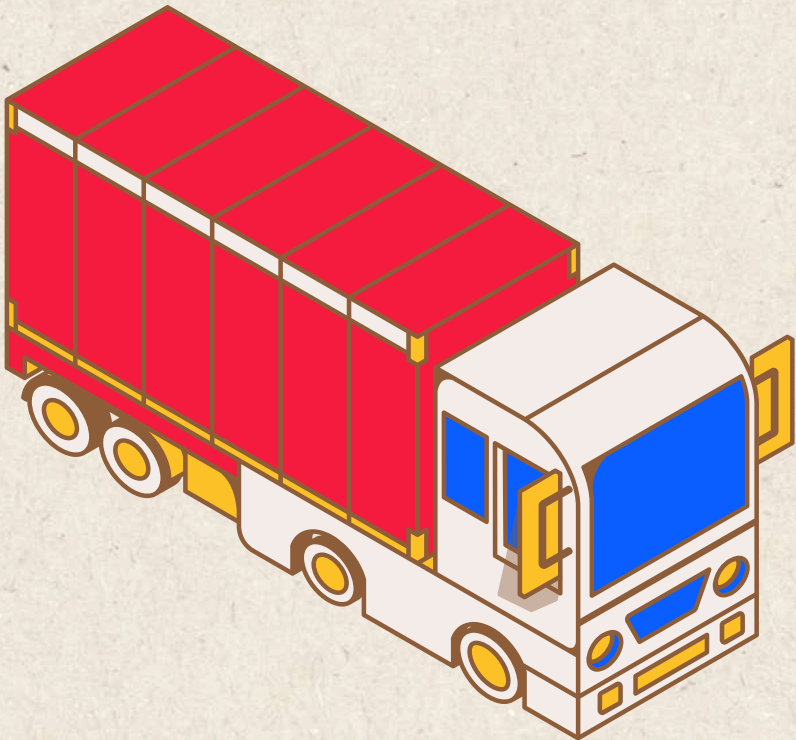
- Show transportation modes ordered by total logistics cost

Code

```
SELECT Transportation_modes, SUM(Costs) AS TotalCost
FROM a1.logistics
GROUP BY Transportation_modes
ORDER BY TotalCost DESC;
```

Output

Result Grid			Filter Rows:
	Transportation_modes	TotalCost	
▶	Road	16048.193638899997	
	Rail	15168.931558400001	
	Air	14604.5274976	
	Sea	7102.925520499999	



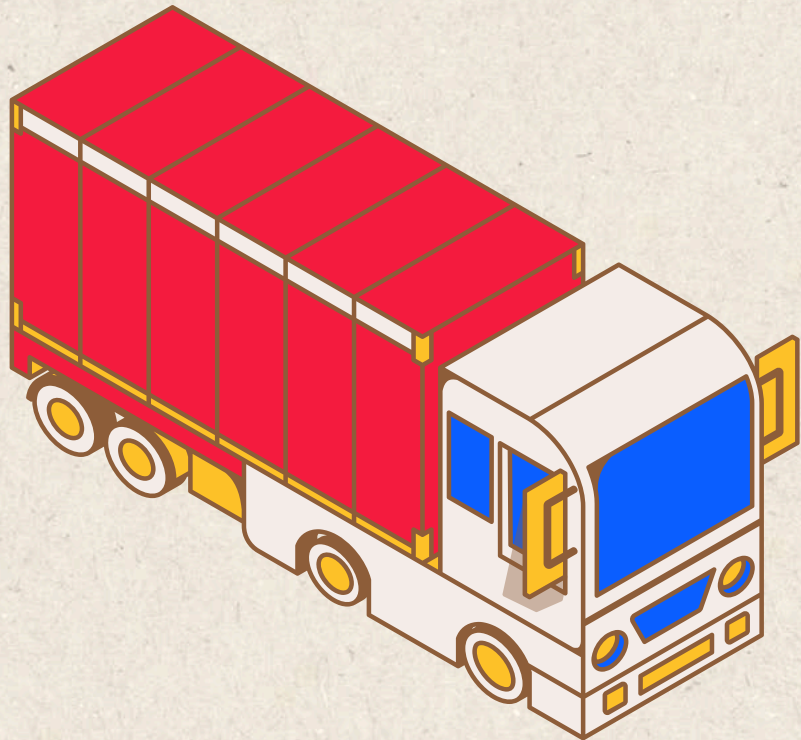
- Show skincare products sold (SKU + units + revenue)

Code

```
SELECT p.Product_type, s.SKU, s.Number_of_products_sold, s.Revenue_generated
FROM a1.sales_data s
JOIN a1.product_table p ON s.SKU = p.SKU
WHERE p.Product_type = 'skincare';
```

Output

	Product_type	SKU	Number_of_products_sold	Revenue_generated
▶	skincare	SKU1	736	7460.900065
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	skincare	SKU9	980	4971.145988
	skincare	SKU10	996	2330.965802
	skincare	SKU11	960	6099.944116
	skincare	SKU13	249	4052.738416
	skincare	SKU14	562	8653.570926



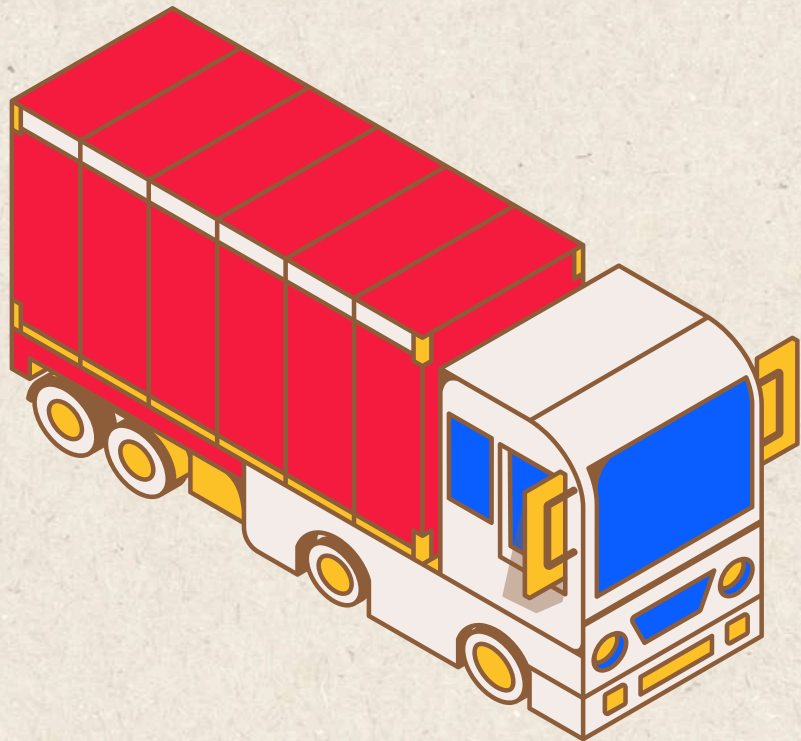
- Show revenue and units sold per product type

Code

```
SELECT p.Product_type, SUM(s.Revenue_generated) AS TotalRevenue, SUM(s.Number_of_products_sold) AS TotalUnits
FROM a1.sales_data s
JOIN a1.product_table p ON s.SKU = p.SKU
GROUP BY p.Product_type;
```

Output

	Product_type	TotalRevenue	TotalUnits
▶	haircare	174455.390606	13611
	skincare	241628.16213300003	20731
	cosmetics	161521.26600100004	11757



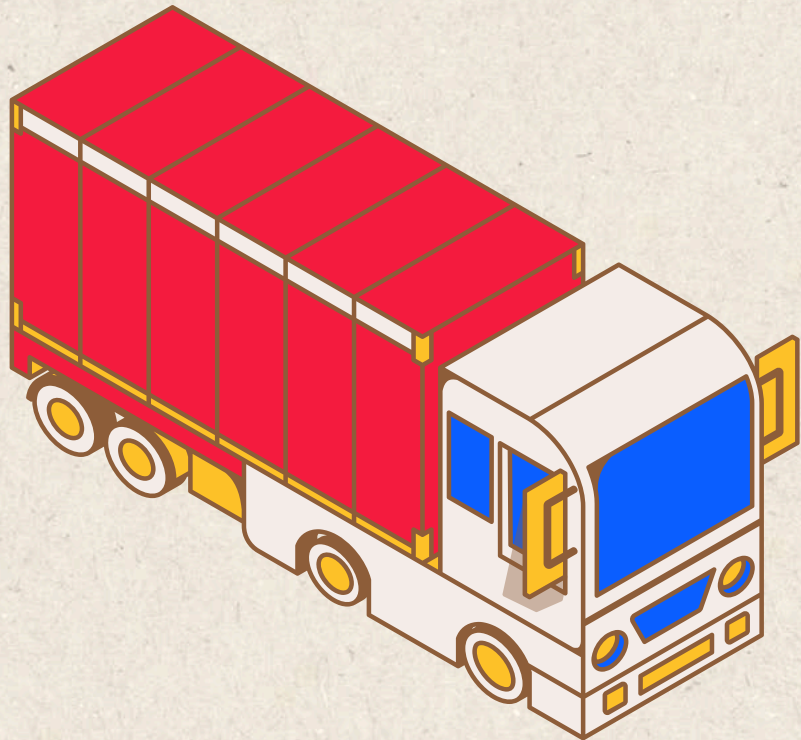
- Show revenue and units sold per product type

Code

```
SELECT
  p.Product_type,
  SUM(s.Revenue_generated) AS TotalRevenue,
  SUM(s.Number_of_products_sold) AS TotalUnits
FROM a1.sales_data s
JOIN a1.product_table p
  ON s.SKU = p.SKU
GROUP BY p.Product_type;
```

Output

	Product_type	TotalRevenue	TotalUnits
▶	haircare	174455.390606	13611
	skincare	241628.16213300003	20731
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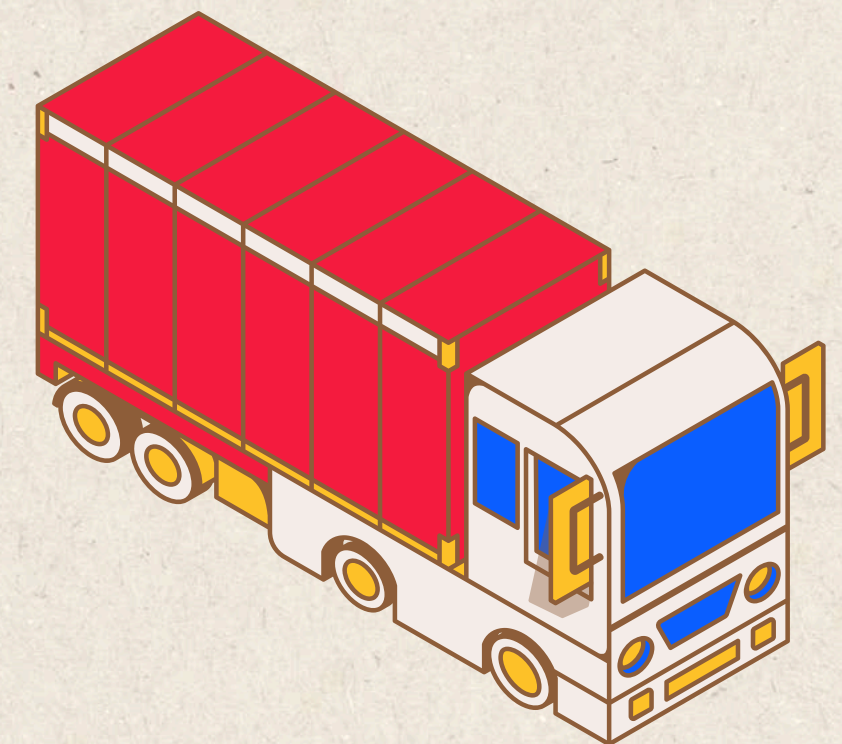
- Show average lead time of all suppliers

Code

```
SELECT p.Product_type, SUM(s.Revenue_generated) AS TotalRevenue, SUM(s.Number_of_products_sold) AS TotalUnits
FROM a1.sales_data s
JOIN a1.product_table p ON s.SKU = p.SKU
GROUP BY p.Product_type;
```

Output

	AvgLeadTime
▶	17.0800



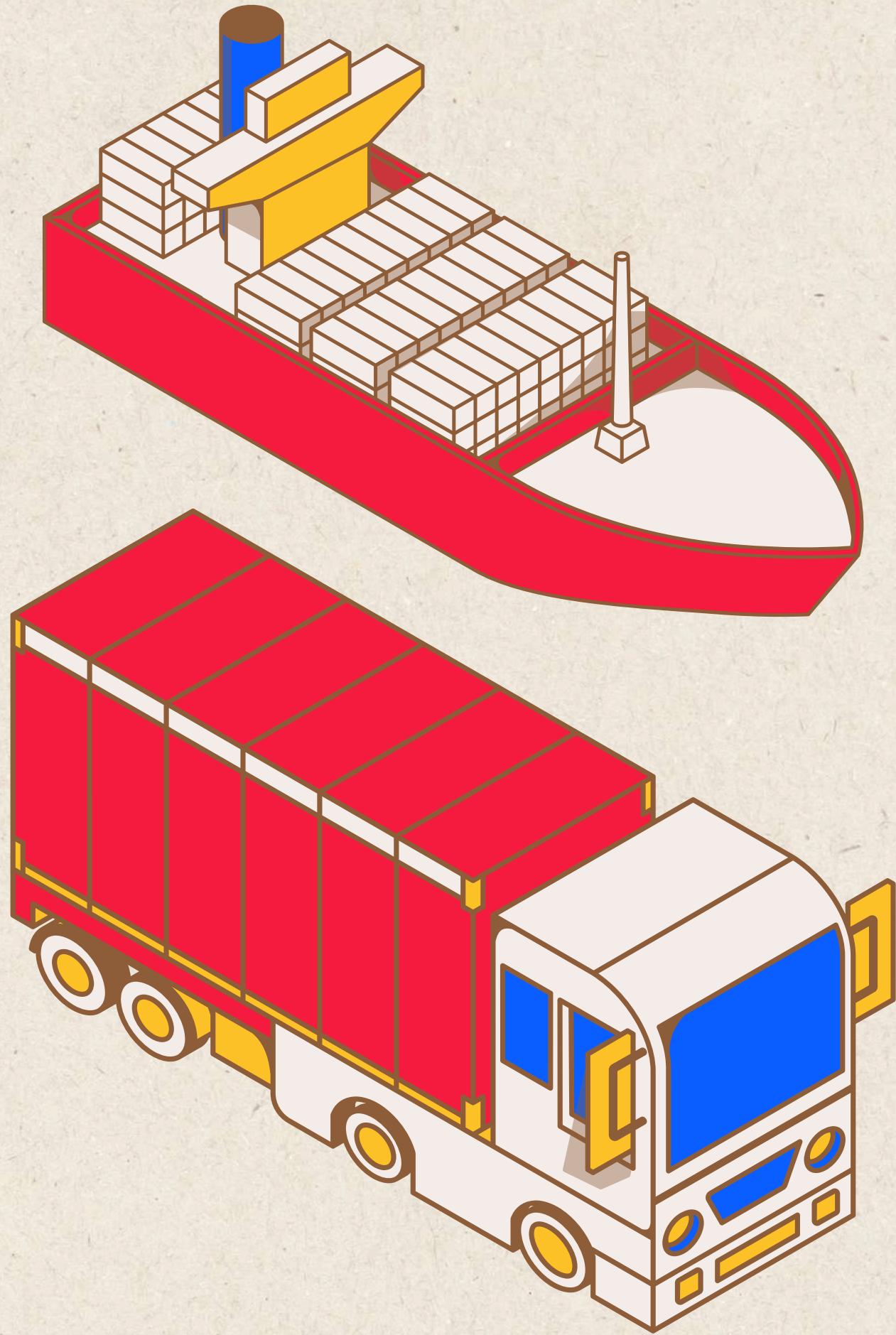
Key Insights

- Skincare and similar product categories show high revenue potential.
- A few suppliers contribute the most to total production — indicating dependency risk.
- Some transportation modes incur higher costs, suggesting optimization opportunities.
- Manufacturing defect rates vary significantly — continuous monitoring is required.
- Certain products consistently outperform others, useful for future marketing focus.

Conclusion

- The SQL-based analysis provides a 360° view of the company's operations, helping decision-makers:
- Optimize supplier relationships and logistics strategies.
- Focus production on high-demand, high-revenue products.
- Improve manufacturing efficiency by reducing defect rates.
- Enhance profitability by understanding cost and revenue distribution.
- This project demonstrates how SQL can be used not just for data retrieval but for strategic business intelligence and performance optimization.

Thank You So Much!



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